

CASE STUDY

Windows Failed Logon Investigation & Detection

Hands-on SOC L1 lab using Windows logs and Splunk

Category

Splunk · Windows Security · Event Viewer · Detection

Status

Completed

Primary event

Event ID 4625 — Failed Logon

Environment

Kali Linux VM + Splunk Enterprise (local trial)

Overview

A hands-on SOC L1 lab that mirrors a junior analyst workflow: generate a Windows failed-logon event, analyze it in Event Viewer and PowerShell, ingest it into Splunk, and build a threshold-based detection with a scheduled alert.

Core workflow

- Create a local Windows test account named soclab.
- Generate Event ID 4625 by using runas with an incorrect password.
- Review the event in Event Viewer, then re-read it with Get-WinEvent in PowerShell.
- Export the last 50 events to failed_logons.csv and upload the file into Splunk through Add Data → Upload.
- Build a detection that counts failed attempts within 5-minute windows.
- Save the search as a scheduled alert named Repeated Failed Logons - Windows.

Manual investigation

Key fields were reviewed in Event Viewer, including TargetUser, Logon Type 2, and Status/SubStatus. The same event was then read again with Get-WinEvent in PowerShell to confirm that the same security data is accessible from more than one tool, followed by an XML view to expose structured fields suitable for SIEM analysis.

Ingestion / operational context

The data did not arrive through live forwarding. Instead, the last 50 events were exported to a CSV file and uploaded into Splunk. This distinction matters because it places the project in the investigation-and-detection scope rather than log-collection engineering with a Universal Forwarder.

Detection logic

The query groups failed attempts into 5-minute buckets and returns only windows with 5 or more failed attempts for the same user/source/logon type. Two windows were returned with 5 and 6 attempts respectively, then the search was saved as a scheduled alert that runs every 5 minutes over the last 5 minutes of data.

SPL Query

```
index=main source="failed_logons.csv" EventID=4625
| bin _time span=5m
| stats count as FailedAttempts by _time TargetUser SourceIP LogonType Status
SubStatus
| where FailedAttempts >= 5
| sort - _time
```

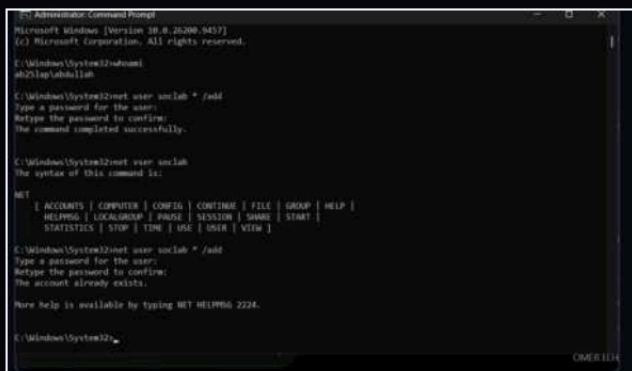
What I learned

Understanding Event ID 4625
Using Event Viewer and PowerShell together
Using Splunk Add Data for CSV ingestion
Building threshold-based SPL detections
Saving a scheduled alert
Treating findings as signals, not conclusions

Important limitations

- The result is a signal worth investigating, not automatic proof of brute force.
- Ingestion was CSV-based rather than live forwarding from Windows.
- The project demonstrates the Event → Investigation → Detection → Alert workflow in a local training lab.

Selected evidence



```
Microsoft Windows [version 10.0.20290.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cmd /c net user socslab /add

Type a password for the user:
Retype the password to confirm:
The command completed successfully.

C:\Windows\System32>net user socslab
The syntax of this command is:

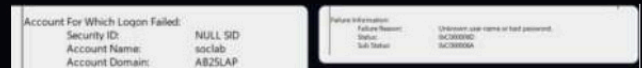
NET
[ ACCOUNTS | COMPUTER | CONFIG | CONTINUE | FILE | GROUP | HELP |
  HELPMSG | LOCALGROUP | PHONE | SESSION | SHARE | START |
  STATISTICS | STOP | TIME | USER | USER | VIEW ]

C:\Windows\System32>net user socslab * /add
Type a password for the user:
Retype the password to confirm:
The account already exists.

More help is available by typing NET HELPMSG 2224.

C:\Windows\System32>
```

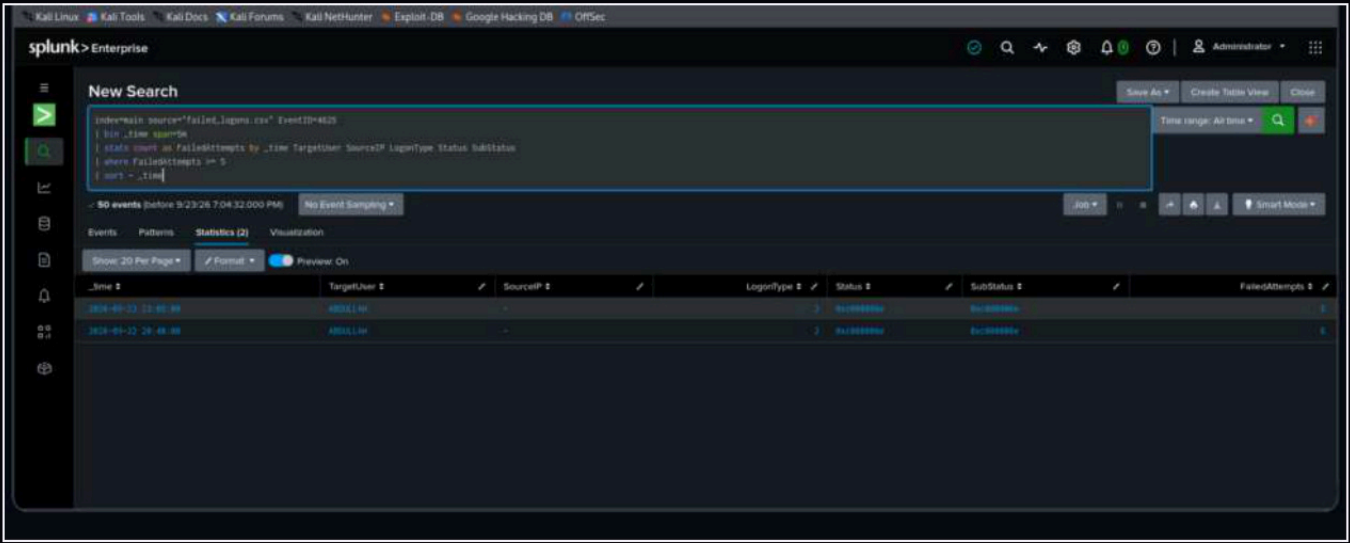
Evidence 1



Account For Which Logon Failed:		Failure Information	
Security ID:	NULL SID	Failure Reason:	Unknown user name or bad password.
Account Name:	socslab	Status:	0xC000006D
Account Domain:	AB2SLAP	Sub Status:	0xC000000A

Evidence 2

Additional evidence



This document is a portfolio case-study summary prepared for professional presentation. It documents analysis and detection work performed on training data / lab environments only.